

A proof of the Linear Hadwiger's conjecture.

Abstract

We prove that there is an absolute constant C such that every graph with no K_t minor has chromatic number at most Ct .

We first show that

$$\chi(G) \leq 4h(G) + \varepsilon|G| + A_\varepsilon,$$

where $h(G)$ is the largest order of a complete minor and $\log A_\varepsilon = O(\varepsilon^{-2} \log(1/\varepsilon))$ as $\varepsilon \rightarrow 0$. We then show that a local linear coloring bound for graphs of order at most $t(\log t)^\alpha$, for any fixed $\alpha > 0$, implies such a bound at exponent $4\alpha/3$. Iterating this improvement and applying the small-graph reduction of Delcourt and Postle proves Linear Hadwiger's conjecture.

1 Introduction

All graphs are finite and simple. A graph H is a minor of a graph G if a graph isomorphic to H can be obtained from a subgraph of G by contracting edges. We write K_t for the complete graph on t vertices. In 1943, Hadwiger [6] conjectured that every K_t -minor-free graph is $(t-1)$ -colorable. The conjecture holds for $t \leq 6$. Wagner's theorem [18] makes the case $t = 5$ equivalent to the Four Color Theorem, and Robertson, Seymour, and Thomas [14] proved the case $t = 6$ by a further reduction to that theorem.

The linear weakening asks whether an absolute constant C suffices to color every K_t -minor-free graph with at most Ct colors. This question appears explicitly in Reed and Seymour [13, p. 148] and is formulated as *Linear Hadwiger's conjecture* in Norin, Postle, and Song [12, Conjecture 1.2] and Delcourt and Postle [1, Conjecture 1.2]. Reed and Seymour proved a fractional version: the fractional chromatic number of every K_t -minor-free graph is at most $2(t-1)$ [13, Theorem 1.3].

For ordinary coloring, Kostochka [8] and Thomason [17] independently obtained the bound $O(t\sqrt{\log t})$ in the 1980s. Their results bound the average degree of every K_t -minor-free graph, and hence give upper bounds of this order for its degeneracy and chromatic number. The density bound is sharp up to a constant factor, so a better coloring bound requires more than a stronger bound on degeneracy. Norin, Postle, and Song [12, Theorem 1.4] broke this barrier by proving $O(t(\log t)^{1/4+\delta})$ -colorability for every fixed $\delta > 0$.

Delcourt and Postle [1, Theorem 1.5] subsequently obtained $O(t \log \log t)$ -colorability. They also reduced the linear conjecture to graphs of order $O(t(\log t)^4)$ and proved it for graphs excluding any fixed complete graph as a subgraph [1, Theorem 1.6 and Corollaries 1.7–1.10]. More recently, Liu and Luo [10, Theorem 1.1] improved the general bound to $O(t \log \log \log t)$. Lin [9, Theorem 1.4 and p. 3] improved the bound on the order of the highly connected subgraphs used in the Delcourt–Postle reduction, lowering its order cutoff to $O(t(\log t)^2)$. We prove the linear conjecture.

Theorem 1. *There is an absolute constant C such that, for every integer $t \geq 2$, every K_t -minor-free graph G satisfies $\chi(G) \leq Ct$.*

We write $|G|$, $\chi(G)$, $\alpha(G)$, and $h(G)$ for the order, chromatic number, independence number, and largest order of a complete minor of G , respectively. These parameters are zero for the empty graph. A stable set is an independent set. All logarithms are natural.

The proof has two parts. First, we approximate ordinary coloring by fractional coloring after adding a graph of bounded maximum degree. A weighted matching contraction controls the resulting fractional chromatic number in terms of the complete minors of the original graph. This yields a local linear coloring bound at a positive logarithmic order exponent. Second, we improve that exponent by packing connected induced bipartite subgraphs and localizing chromatic number near a short induced path. An adaptation of the recursive assembly of Delcourt and Postle completes this improvement. Repeating it reaches the local range required by their small-graph reduction. Section 2 states the two auxiliary theorems and derives Theorem 1 from them; their proofs occupy Sections 3 and 4.

AI disclosure

The proof presented in this paper was developed by OpenAI’s ChatGPT model Astra 6, following research directions and feedback supplied by the authors. The authors suggested several possible approaches, but almost none of their specific proof ideas were retained in the final argument. (Their suggestion to further exploit Liu and Luo’s ideas of contractions involving unusually large independent sets in vertex neighbourhoods is reflected in the packing argument in Section 4.1.)

The project began with the authors asking the model to prove the very dense case of Linear Hadwiger’s conjecture: there exists an absolute constant $C > 0$ such that, for every $C' > 0$, there exists $t_0 > 0$ such that, for every integer $t \geq t_0$, every graph G satisfying $|V(G)| \leq C't$ and $\chi(G) > Ct$ contains a K_t minor. After the model succeeded, the authors asked it to make the dependence of t_0 on C' explicit. This request yielded a proof of linear Hadwiger’s conjecture for graphs G with $h(G) = t$ and $|V(G)| \leq t(\log t)^{1/2-o(1)}$, while by Delcourt-Postle result establishing it for graphs with $|V(G)| \leq t(\log t)^{4+o(1)}$ suffices.

Several approaches to were tried at this point. We asked the model, in particular, to find an original idea to bridge the gap between the two bounds, rather than trying to tighten them. This led to the bootstrap argument used in the present paper.

1.1 Provenance of the proof ideas

The hypergraph matching argument in Section 3 follows the Rödl nibble method [15] and Pippenger’s almost-perfect-matching theorem for nearly regular hypergraphs with small codegrees; see [2, Theorem 1.1] for the latter statement. We give an explicit proof because the dependence on the uniformity is needed when choosing ε . Independent vertex deletions that equalize survival probabilities occur in Kang, Kühn, Methuku, and Osthus [7, Section 5.1] and Gould and Kelly [4, Section 6.1]. Our implementation uses Poisson marks and controls the total degree deficit after trimming. We apply the resulting matching estimate to an auxiliary hypergraph whose edges encode stable sets and whose additional vertices control the number of colors.

The passage from this rounding estimate to the quantitative coloring bound uses two further constructions proved here. A linear program with constraints on the weights of pairs yields a bounded-degree augmentation whose fractional chromatic number approximates the original chromatic number. A weighted matching contraction then transfers fractional-coloring weights to a minor of the original graph. Reed and Seymour’s theorem [13, Theorem 1.3], applied to that minor, supplies the complete-minor bound. Thus their fractional theorem is a direct input, while the augmentation and matching contraction provide the additional comparison with ordinary coloring.

The factor-two lifting of a quotient coloring through connected bipartite pieces already appears in Reed and Seymour’s discussion of chordal decompositions [13, p. 149]. Liu and Luo [10, Lemma 3.1] use a related contraction of induced stars with independent centers, for which the coloring loss is only one additional color. In Section 4, a maximal packing of connected induced bipartite subgraphs of prescribed order leaves a remainder with no large such subgraph. We then adapt the induced-path construction of Gyárfás [5, proof of Theorem 2.4]: following a component of large chromatic number localizes the chromatic obstruction in the neighborhood of a short path. The absence of large induced stars bounds the independence number of that neighborhood, and the Reed–Seymour bound converts this into the order bound needed for the local coloring hypothesis.

The outer recursion is a direct adaptation of Delcourt and Postle [1, Definition 2.4 and Section 7]. We use their chromatic separability framework and their assembly of a complete minor from three smaller models, together with their notion of wovenness and linkage rerouting lemma [1, Definition 5.12 and Lemma 5.15]. Our fan lemma follows from their redundant form of Menger’s theorem [1, Lemma 5.3]. The routing hub is obtained from the connectivity theorem of Girão and Narayanan [3], in the quantitative form recorded in [1, Theorem 5.1]; Nguyen’s stronger theorem [11, Theorem 1.2] also suffices. Choosing the hub inside a subgraph of prescribed chromatic number controls the chromatic cost of its deletion. This permits the recursion to use the separation supplied by the packing and path argument. The remaining linkage and rooted-minor inputs are the Thomas–Wollan theorem [16, Corollary 1.3] and Kawarabayashi’s lemma in the form stated in [1, Lemma 5.14].

2 Two auxiliary theorems and the main deduction

Our first auxiliary theorem supplies the initial local coloring bound.

Theorem 2. *For every graph G and every $0 < \varepsilon \leq 1$,*

$$\chi(G) \leq 4h(G) + \varepsilon|G| + A_\varepsilon, \quad A_\varepsilon = (100/\varepsilon)^{2000/\varepsilon^2}. \quad (1)$$

For $\alpha > 0$, say that a *local linear bound holds at exponent α* if there are constants $D \geq 1$ and t_0 such that, for every integer $t \geq t_0$,

$$G \text{ is } K_t\text{-minor-free}, \quad |G| \leq t(\log t)^\alpha \implies \chi(G) \leq Dt.$$

The constants may depend on the fixed exponent. The next theorem improves any such bound to a larger exponent.

Theorem 3. *If a local linear bound holds at exponent $\alpha > 0$, then a local linear bound holds at exponent $4\alpha/3$.*

We prove Theorems 2 and 3 in Sections 3 and 4, respectively. We first derive Theorem 1 from them. The external input for this deduction is the following small-graph reduction.

Theorem 4 (Delcourt–Postle, [1, Theorem 1.6]). *There is an absolute integer $C_{\text{DP}} \geq 1$ such that, for every integer $t \geq 3$ and every K_t -minor-free graph G ,*

$$\chi(G) \leq C_{\text{DP}}t(1 + f(G, t)),$$

where

$$f(G, t) = \max \left(\{0\} \cup \left\{ \frac{\chi(H)}{a} : \begin{array}{l} H \subseteq G \text{ is } K_a\text{-minor-free,} \\ a \in \mathbb{Z}, \quad t/\sqrt{\log t} \leq a \leq t, \\ |H| \leq C_{\text{DP}}a(\log a)^4 \end{array} \right\} \right).$$

We will also use the following explicit complete-minor density bound.

Theorem 5 (Kostochka–Thomason, [1, Theorem 4.3]). *Let $r \geq 2$ be an integer. Every nonempty graph H with $|E(H)|/|H| \geq 30r\sqrt{\log r}$ contains a K_r minor.*

Consequently every K_r -minor-free graph satisfies

$$\chi(H) \leq 60r\sqrt{\log r} + 1. \quad (2)$$

Indeed, the bound is immediate for the empty graph. Otherwise, an s -chromatic graph has a vertex-critical subgraph C with minimum degree at least $s - 1$, whence

$$\frac{|E(C)|}{|C|} \geq \frac{s - 1}{2}.$$

Theorem 5 applies if the displayed coloring bound fails.

Proof of Theorem 1. Put $\varepsilon = 1/\lceil 2(\log t)^{1/3} \rceil$. For all sufficiently large t , this lies in $(0, 1]$, and

$$\log A_\varepsilon = \frac{2000}{\varepsilon^2} \log \frac{100}{\varepsilon} = 8000(\log t)^{2/3} \log(200(\log t)^{1/3}) = o(\log t).$$

Consequently $A_\varepsilon \leq t/2$ eventually. If G is K_t -minor-free and $|G| \leq t(\log t)^{1/3}$, then $h(G) \leq t - 1$ and $\varepsilon|G| \leq t/2$. Theorem 2 gives $\chi(G) \leq 4(t - 1) + t/2 + t/2 < 5t$.

Thus a local linear bound holds at exponent $1/3$. Applying Theorem 3 nine times, each time multiplying the exponent by $4/3$, gives a local linear bound at exponent $\alpha = (4/3)^9/3 > 4$. Let D, t_0 be the constants in this bound. For all sufficiently large a ,

$$C_{\text{DP}}(\log a)^4 \leq (\log a)^\alpha.$$

For sufficiently large t , every integer a in the window of Theorem 4 satisfies this inequality and $a \geq t_0$. Thus $f(G, t) \leq D$ and $\chi(G) \leq C_{\text{DP}}(1 + D)t$. The finitely many smaller values of t are absorbed into the constant using (2). The case $t = 2$ is immediate. $\square$

3 Proof of the quantitative coloring bound

We prove Theorem 2 by combining three ingredients. An explicit hypergraph-matching estimate rounds a fractional coloring whose pair loads are small. Linear-programming duality then produces a bounded-degree augmentation whose fractional chromatic number approximates $\chi(G)$. Finally, a weighted matching contraction bounds that fractional chromatic number by $4h(G)$, up to an additive term depending only on the independence number and the augmentation degree.

A fractional coloring assigns nonnegative weights x_S to the nonempty stable sets of a graph, with

$$\sum_{S \ni v} x_S \geq 1 \quad (v \in V(G)).$$

Its cost is $\sum_S x_S$, and the minimum cost is the fractional chromatic number $\chi_f(G)$. Linear-programming duality gives

$$\chi_f(G) = \max \left\{ \sum_v y_v : y_v \geq 0, \quad \sum_{v \in S} y_v \leq 1 \text{ for every stable set } S \right\}. \quad (3)$$

In particular, every dual-feasible weight satisfies $y_v \leq 1$. Reed and Seymour [13, Theorem 1.3] proved that every graph J satisfies

$$\chi_f(J) \leq 2h(J). \quad (4)$$

The empty graph satisfies this inequality directly.

3.1 An explicit almost-perfect matching lemma

A multihypergraph is r -uniform if every edge is a set of r vertices; repeated copies of an edge are allowed. The degree of a vertex is the number of incident edge copies, and the codegree of two distinct vertices is the number of edge copies containing both. A matching consists of pairwise disjoint edges.

Lemma 6. *Let $r \geq 2$ be an integer, let $0 < \xi \leq 1$, and put*

$$\mu = \left(\frac{\xi}{100r} \right)^{20r}.$$

Every finite r -uniform multihypergraph with positive maximum degree D_0 , minimum degree at least $(1 - \mu)D_0$, and maximum codegree at most μD_0 has a matching covering at least a $(1 - \xi)$ -fraction of its vertices.

Proof. We equalize vertex survival probabilities and control the total degree deficit after each round. Write $x_+ = \max\{x, 0\}$. Let H_0 be such a hypergraph, let $N = |V(H_0)|$, and let B be its maximum codegree. Since H_0 has an edge and $r \geq 2$, we have $B \geq 1$ and hence $D_0 \geq 1/\mu$.

Set

$$L = \log(4/\xi), \quad a = \frac{\xi}{100rL}, \quad q = e^{-a}, \quad T = \lceil L/a \rceil, \quad D_{i+1} = \lfloor q^{r-1} D_i \rfloor.$$

We first record bounds ensuring that all T rounds are well defined. We have $1 \leq L \leq 4/\xi$, $ar \leq 1/100$, and $aT \leq 2L$. Put

$$b = 2\mu(4/\xi)^{2r}.$$

The definition of μ gives

$$\sqrt{b} = \frac{\sqrt{2} 4^r \xi^{9r}}{(100r)^{10r}} \leq \frac{\xi^5}{10^8 r^3}. \quad (5)$$

Indeed, $9r \geq 5$ and $\sqrt{2} 4^r / (100r)^{10r} \leq 1/(10^8 r^3)$ for $r \geq 2$. Since $L \leq 4/\xi$, we also have $a \geq \xi^2/(400r)$ and hence

$$b \leq \frac{\xi^{10}}{10^{16} r^6} \leq \frac{\xi^2}{1600r} \leq \frac{a}{4}.$$

The definition of μ also gives $\mu \leq \xi^2/800$.

The geometric sum of the rounding errors in D_i is at most $1/(1 - q^{r-1}) \leq 2/a$, since $r \geq 2$ and $1 - e^{-a} \geq a/2$. Thus, for $0 \leq i \leq T$,

$$\begin{aligned} D_i &\geq D_0 e^{-(r-1)ai} - 2/a \\ &\geq D_0 (\xi/4)^{2r} - 2/a \geq \frac{1}{2} D_0 (\xi/4)^{2r} \geq 4/a. \end{aligned}$$

Here $D_0 (\xi/4)^{2r} \geq 2/b \geq 8/a$. Consequently

$$B/D_i \leq b, \quad D_{i+1} \geq D_i/2, \quad D_{i+1} \geq q^r D_i. \quad (6)$$

For the last inequality, note that

$$q^{r-1} D_i (1 - q) \geq e^{-1/100} (4/a) (a/2) > 1,$$

so $\lfloor q^{r-1} D_i \rfloor \geq q^r D_i$. These bounds also imply $ar^2 B/D_i \leq 1/2$.

We construct a sequence H_i of residual hypergraphs with maximum degree at most D_i and codegree at most B . Write

$$n_i = |V(H_i)|, \quad S_i = n_i D_i - r|E(H_i)|.$$

The nonnegative quantity S_i is the total degree deficit. Conditional on H_i , attach to every edge an independent Poisson variable of mean a/D_i . Call the edge marked when this variable is positive. Independently attach to each vertex v a private Poisson variable of mean

$$a \left(1 - \frac{d_{H_i}(v)}{D_i}\right).$$

Keep precisely the vertices whose private variable and all incident edge variables are zero. Add to the matching every marked edge disjoint from all other marked edges. All endpoints of these matching edges are deleted, so the matchings obtained in different rounds are disjoint. Private variables do not affect acceptance of an isolated marked edge.

Let J_i be induced by the surviving vertices. Delete edges of J_i until its maximum degree is at most D_{i+1} , and call the result H_{i+1} . This trimming does not delete vertices or increase codegrees.

Every vertex survives with probability exactly q , so

$$\mathbb{E}[n_{i+1} \mid H_i] = qn_i, \quad \mathbb{E}n_i = q^i N. \quad (7)$$

More generally, if U is a set of k vertices, independence of the Poisson variables gives

$$\mathbb{P}(U \text{ survives} \mid H_i) = q^k \exp \left(\frac{a}{D_i} \sum_{e \in E(H_i)} (|e \cap U| - 1)_+ \right). \quad (8)$$

The sum in the exponent is between zero and $\binom{k}{2}B$, since $(j-1)_+ \leq \binom{j}{2}$. In particular, every edge survives with probability at least q^r .

We next bound the expected number of trimmed edges. Fix $v \in V(H_i)$ and condition on its survival. Let $X_v = d_{J_i}(v)$ under this conditioning. Dividing (8) for an edge containing v by q yields

$$\mathbb{E}X_v \leq d_{H_i}(v)q^{r-1} + 2ar^2B.$$

We used $e^x \leq 1 + 2x$ for $0 \leq x \leq 1/2$ and $d_{H_i}(v) \leq D_i$.

For completeness, we bound the conditional variance directly. In the expansion of $\text{Var } X_v$, the number of ordered pairs of incident edges sharing a vertex other than v , including diagonal pairs, is at most

$$\sum_{u \neq v} d_{H_i}(u, v)^2 \leq B \sum_{u \neq v} d_{H_i}(u, v) \leq rBD_i.$$

Their individual covariance is at most one. For any other pair e, f , the sets $e - v$ and $f - v$ are disjoint. Conditional on v surviving, its incident edge variables and its private variable are forced to be zero; all other variables remain independent. The covariance of the survival indicators of these two sets is

$$P_e P_f (e^\lambda - 1),$$

where P_e, P_f are their conditional survival probabilities and λ is the sum of the means of the remaining Poisson variables affecting both sets. Private variables cannot affect both. The codegree bound gives $\lambda \leq ar^2B/D_i$, and hence the covariance is at most $2ar^2B/D_i$. There are at most D_i^2 such ordered pairs. Therefore

$$\text{Var } X_v \leq rBD_i + 2ar^2BD_i \leq 4r^2BD_i.$$

It follows from Cauchy–Schwarz and the rounding in D_{i+1} that

$$\mathbb{E}(X_v - D_{i+1})_+ \leq 2r\sqrt{BD_i} + 2ar^2B + 1.$$

All expectations and variances in this paragraph are conditional on H_i and survival of v .

Let F_i be the number of edges deleted in trimming. At each deletion, choose an edge incident with a vertex whose degree exceeds D_{i+1} . Each deletion reduces the sum of positive excess degrees by at least one. Thus

$$\mathbb{E}[F_i \mid H_i] \leq qn_i(2r\sqrt{BD_i} + 2ar^2B + 1).$$

Since $qD_{i+1} \leq q^r D_i$ and every edge survives with probability at least q^r , we obtain

$$\mathbb{E}[S_{i+1} \mid H_i] \leq q^r S_i + rqn_i(2r\sqrt{BD_i} + 2ar^2B + 1). \quad (9)$$

Let W_i be the number of vertices deleted in round i that are not covered by its accepted edges. Every such vertex either has a positive private variable or belongs to a marked edge intersecting another marked edge. The expected first count is at most aS_i/D_i . There are at most $|E(H_i)|rD_i$ ordered pairs of distinct intersecting edges; each pair is marked with probability at most $(a/D_i)^2$. Charging at most r vertices to its first edge and using $r|E(H_i)| \leq n_i D_i$ gives

$$\mathbb{E}[W_i \mid H_i] \leq aS_i/D_i + ra^2n_i. \quad (10)$$

It remains to sum these losses. Put $Z_i = \mathbb{E}S_i/D_i$. By (6) and (9), using $B \geq 1$, $b \leq 1$, and $ar \leq 1$,

$$\begin{aligned} Z_{i+1} &\leq Z_i + \frac{2rN}{D_i}(2r\sqrt{BD_i} + 2ar^2B + 1) \\ &\leq Z_i + N(4r^2\sqrt{b} + 4ar^3b + 2rb) \\ &\leq Z_i + 10r^2N\sqrt{b}. \end{aligned}$$

Here $D_{i+1} \geq D_i/2$ and $\mathbb{E}n_i \leq N$ give the first inequality, while $B/D_i \leq b$ and $1/D_i \leq B/D_i$ give the second. Initially $Z_0 \leq \mu N$, so $Z_i \leq \mu N + 10ir^2N\sqrt{b}$. Summing (10), and using $T \leq 2L/a$, gives

$$\begin{aligned} \frac{1}{N}\mathbb{E}\sum_{i < T} W_i &\leq aT\mu + 10aT^2r^2\sqrt{b} + ra^2T \\ &\leq 2L\mu + \frac{4000r^3L^3}{\xi}\sqrt{b} + \frac{\xi}{50} < \frac{\xi}{25}. \end{aligned}$$

For the last inequality, the first term is at most $\xi/100$, and (5) and $L \leq 4/\xi$ bound the second by $256000\xi/10^8 < \xi/100$. Finally (7) gives $\mathbb{E}n_T = q^T N \leq \xi N/4$. The vertices not covered by the union of the accepted matchings are precisely those counted by $n_T + \sum_{i < T} W_i$. Their expected number is less than ξN , so some outcome proves the lemma. $\square$

3.2 Fractional coloring and low-codegree rounding

A fractional coloring can be rounded efficiently when no pair of vertices receives much common weight. The color tokens in the construction below control the number of stable sets in the resulting coloring.

Lemma 7. *Let $m \geq 1$ be an integer and let $0 < \gamma \leq 1$. Set*

$$r = m + 2, \quad \mu = \left(\frac{\gamma}{400r^2}\right)^{20r}, \quad n_0 = \left\lceil 10^6 r^4 \mu^{-4} + 2/\gamma \right\rceil.$$

Suppose G has $n \geq n_0$ vertices, $\alpha(G) \leq m$, and a fractional coloring of cost τ satisfying

$$\sum_{S \ni v} x_S = 1, \quad \sum_{S \supseteq \{u,v\}} x_S \leq \mu/4 \quad (u \neq v).$$

Then $\chi(G) \leq \tau + \gamma n$.

Proof. Exact vertex loads give

$$n = \sum_S |S| x_S, \quad n/m \leq \tau \leq n, \quad x_S \leq 1.$$

Take three disjoint vertex sets: $V(G)$, a set P of $p = \lceil \tau \rceil$ color tokens, and a set Z of $z = \lceil (r-1)\tau - n \rceil$ dummy vertices. For every nonempty stable set S , token $c \in P$, and set $T \subseteq Z$ of size $r-1-|S|$, assign the r -set $S \cup \{c\} \cup T$ weight

$$\omega(S, c, T) = \frac{x_S}{p \binom{z}{r-1-|S|}}.$$

Since $1 \leq |S| \leq m$, each edge contains between one and m dummy vertices. Also

$$(r-1)\tau - n \geq \tau \geq n/m, \quad n + p + z \leq r\tau + 2 \leq rn + 2.$$

Thus $p, z \geq n/m$; the assumed lower bound on n also gives $z \geq m+1$.

The weighted vertex degrees are 1 at original vertices, τ/p at tokens, and $((r-1)\tau - n)/z$ at dummy vertices. Both τ and $(r-1)\tau - n$ are at least n/m , while each ceiling adds less than one to its denominator. Hence each of these degrees is at least

$$\frac{n/m}{n/m+1} \geq 1 - \frac{m}{n},$$

and all are at most one. The weighted codegree of two original vertices is at most $\mu/4$. For an original vertex and a token it is $1/p$; for an original vertex and a dummy it is at most m/z ; for a token and a dummy it is $((r-1)\tau - n)/(pz)$; and for two dummies it is at most

$$\frac{m(m-1)\tau}{z(z-1)}.$$

Two tokens have codegree zero. For the dummy–dummy bound, for example, $z \geq n/m$, $z-1 \geq n/(2m)$, and $\tau \leq n$ give

$$\frac{m(m-1)\tau}{z(z-1)} \leq \frac{2m^4}{n}.$$

The other displayed codegrees are no larger, so all weighted codegrees other than original–original codegrees are at most $2m^4/n$. Because $\binom{z}{j} \geq z$ for $1 \leq j \leq m$ under our order bound,

$$\omega(S, c, T) \leq \frac{1}{pz} \leq \frac{m^2}{n^2}.$$

Independently sample each edge with probability $n\omega(S, c, T)$, which is at most one. Chernoff bounds show that, with positive probability, all degrees differ from their means by at most $\mu n/8$ and all codegrees are at most $\mu n/2$. Indeed, $n \geq 16m^4/\mu$ gives $2m^4 \leq \mu n/8$, so every codegree has mean at most $\mu n/4$. Since there are at most $rn+2$ vertices, the total failure probability is at most

$$8r^2 n^2 \exp(-\mu^2 n/192) < 1.$$

For the strict inequality, $e^x \geq x^4/24$ and $n \geq 10^6 r^4 \mu^{-4}$ give

$$8r^2 n^2 e^{-\mu^2 n/192} \leq \frac{192^5 r^2}{\mu^8 n^2} \leq \frac{192^5}{10^{12} r^6} < 1.$$

Also $n \geq 8m/\mu$, so the minimum degree is at least $n - m - \mu n/8 \geq (1 - \mu/4)n$, while the maximum degree is at most $(1 + \mu/8)n$. In particular,

$$1 - \mu/4 \geq (1 - \mu)(1 + \mu/8), \quad \mu/2 < \mu(1 - \mu/4),$$

and hence the sampled hypergraph meets both degree hypotheses of Lemma 6.

Apply Lemma 6 with $\xi = \gamma/(4r)$; its parameter is exactly $\mu = (\xi/(100r))^{20r}$. The matching leaves at most $\xi(rn + 2) \leq \gamma n/2$ vertices uncovered. It uses at most p edges because each edge contains a distinct token. The original vertices on each matching edge form a stable set. Coloring the remaining original vertices separately uses at most $p + \gamma n/2 \leq \tau + \gamma n$ colors, since $n \geq 2/\gamma$. $\square$

3.3 A bounded-degree augmentation

We use the pair-load constraints in Lemma 7 to identify a bounded-degree set of nonedges whose addition makes fractional coloring approximate ordinary coloring.

Lemma 8. *Let $m \geq 2$ be an integer and let $0 < \gamma \leq 1$. Let μ, n_0 be the parameters in Lemma 7 for m, γ . Every graph G on $n \geq n_0$ vertices with $\alpha(G) \leq m$ admits a graph F on $V(G)$ such that*

$$E(F) \subseteq E(\overline{G}), \quad \Delta(F) \leq \left\lceil \frac{8 \binom{m}{2}}{\mu \gamma^2} \right\rceil,$$

and

$$\chi_f(G + F) \geq \chi(G) - 3\gamma n.$$

Here $G + F$ is the graph with edge set $E(G) \cup E(F)$.

Proof. Consider the linear program

$$\begin{aligned} \tau &= \min \sum_S x_S, \\ \text{subject to } x_S &\geq 0, \\ \sum_{S \ni v} x_S &\geq 1 \quad (v \in V(G)), \\ \sum_{S \supseteq \{u, v\}} x_S &\leq \mu/4 \quad (u \neq v), \end{aligned} \tag{11}$$

where S ranges over nonempty stable sets. Singleton sets give a feasible solution of cost n , so $\tau \leq n$.

An optimal solution can be chosen with every vertex load exactly one. Indeed, if a vertex is overcovered, transfer appropriate weight from sets containing it to those sets with that vertex removed; delete the transferred weight when the resulting set is empty. This leaves all other vertex loads unchanged and never increases cost or a pair load. Since the original solution was optimal, the resulting feasible solution still has cost τ . Processing all vertices proves the claim. Lemma 7 therefore gives

$$\tau \geq \chi(G) - \gamma n.$$

The dual of (11) is

$$\begin{aligned} \tau &= \max \sum_v a_v - \frac{\mu}{4} \sum_{uv} b_{uv}, \\ \text{subject to } a_v, b_{uv} &\geq 0, \\ \sum_{v \in S} a_v - \sum_{uv \subseteq S} b_{uv} &\leq 1 \quad \text{for every nonempty stable set } S. \end{aligned} \tag{12}$$

Only nonedge pairs need to be included. Fix an optimal dual solution. Singleton constraints give $a_v \leq 1$, and $\tau \geq 0$ gives

$$\frac{\mu}{4} \sum_{uv} b_{uv} = \sum_v a_v - \tau \leq n, \quad \text{so} \quad \sum_{uv} b_{uv} \leq 4n/\mu. \tag{13}$$

Let

$$X = \left\{ v : \sum_u b_{uv} > \frac{8}{\mu\gamma} \right\}.$$

By (13),

$$\frac{8}{\mu\gamma} |X| < \sum_{v \in X} \sum_u b_{uv} \leq 2 \sum_{uv} b_{uv} \leq \frac{8n}{\mu},$$

so $|X| \leq \gamma n$. Define F by adding every nonedge uv with $u, v \notin X$ and

$$b_{uv} \geq \frac{\gamma}{\binom{m}{2}}.$$

Each vertex outside X has total incident penalty at most $8/(\mu\gamma)$, while vertices of X receive no added edges. Consequently

$$\Delta(F) \leq \frac{8\binom{m}{2}}{\mu\gamma^2},$$

which is independent of n .

Put $H = G + F$, and assign weights

$$a'_v = \begin{cases} a_v/(1+\gamma), & v \notin X, \\ 0, & v \in X. \end{cases}$$

If S is stable in H , then $T = S \setminus X$ is stable in G , has at most m vertices, and every pair in T has penalty less than $\gamma/\binom{m}{2}$. The constraint in (12) therefore yields

$$\sum_{v \in T} a_v \leq 1 + \sum_{uv \subseteq T} b_{uv} \leq 1 + \binom{|T|}{2} \frac{\gamma}{\binom{m}{2}} \leq 1 + \gamma.$$

The same conclusion is immediate if T is empty. Hence a' is dual-feasible for fractional coloring of H . Using $\sum_v a_v \geq \tau$, $|X| \leq \gamma n$, and $\tau \leq n$, we obtain

$$\begin{aligned} \chi_f(H) &\geq \frac{\sum_v a_v - |X|}{1+\gamma} \\ &\geq \frac{\tau - \gamma n}{1+\gamma} \\ &\geq \tau - 2\gamma n \\ &\geq \chi(G) - 3\gamma n. \end{aligned}$$

For the penultimate inequality, use $(\tau - \gamma n)/(1 + \gamma) = \tau - \gamma(\tau + n)/(1 + \gamma) \geq \tau - 2\gamma n$, since $\tau \leq n$. This proves the lemma. $\square$

3.4 Robust fractional bound

The augmentation changes the graph whose fractional chromatic number is measured. The next proposition controls this change using the minor number of the original graph. All dependence on the independence number and the degree of the augmentation enters an additive constant.

Proposition 9 (Bounded-degree perturbations). *Let $m \geq 1$ and $d \geq 0$ be integers. Suppose that $\alpha(G) \leq m$ and that F is a graph on $V(G)$ with*

$$E(F) \subseteq E(\overline{G}), \quad \Delta(F) \leq d.$$

Set

$$K = \sum_{j=0}^{2m-1} (d+1)^j.$$

Then

$$\chi_f(G + F) \leq 4h(G) + 2m(K + 1). \quad (14)$$

Lemma 10 (Independent transversal). *Let P be a perfect matching on a finite set U , and let D be a graph on U edge-disjoint from P . If no cycle of $D + P$ contains an edge of P , then D has an independent set containing exactly one endpoint of every edge of P .*

Proof. Every edge of P is a bridge of $D + P$. Contract the connected components of D . No edge of P has both endpoints in one component, since that edge together with a path inside the component would form a cycle. The resulting multigraph is a forest: a cycle, including two parallel edges, would make its matching edges nonbridges in $D + P$.

The degree of the vertex corresponding to a component C of D is exactly $|C|$. Indeed, each vertex of C is incident with one matching edge and every matching edge leaves C . If $U \neq \emptyset$, this forest has a leaf. The corresponding component of D is therefore a singleton $\{x\}$, so x has no neighbor in D .

Select x , and delete x and its matching partner. On the remaining vertices the matching is perfect and the hypothesis about cycles is preserved. Induction supplies an independent transversal of the remaining pairs, and adjoining x proves the lemma. The empty set provides the base case. Notice that cycles consisting only of edges of D cause no difficulty. $\square$

Lemma 11 (Matching with bounded weight loss). *Let $m \geq 1$ and $d \geq 0$ be integers. Let G and D be graphs on the same finite vertex set, with $\alpha(G) \leq m$ and $\Delta(D) \leq d$. Let $w(v) \in [0, 1]$ be vertex weights. Fix an integer $r \geq 1$ and set*

$$B = \sum_{j=0}^r (d+1)^j.$$

There is a matching $M \subseteq E(G)$, with unmatched set X , such that

$$|X| \leq m(B + 1), \quad \sum_{uv \in M} |w(u) - w(v)| \leq m(B + 1), \quad (15)$$

and no cycle of length at most $r + 1$ in $D + M$ contains an edge of M . Moreover, $E(M) \cap E(D) = \emptyset$.

Proof. Maintain a set of unprocessed vertices, initially $V(G)$, and an initially empty matching M . Choose an unprocessed vertex v of maximum weight. Its eligible partners are the other unprocessed vertices u such that $uv \in E(G)$ and the distance from u to v in the current graph $D + M$ is greater

than r . Vertices in different components have infinite distance. If an eligible partner exists, choose one of maximum weight, insert uv into M , and remove both endpoints from the unprocessed set. Otherwise, put v into X and remove it from the unprocessed set. Continue until all vertices have been processed. Throughout this procedure, the graph used to measure distances retains the full vertex set; putting a vertex in X only removes it from consideration as a future partner.

At every stage the graph $D + M$ has maximum degree at most $d + 1$, and its radius- r balls have at most B vertices. Every inserted matching edge is a nonedge of D , because its endpoints have distance greater than $r \geq 1$.

Suppose that the final graph has a cycle of length at most $r + 1$ containing a matching edge. Consider the matching edge of this cycle inserted last. All remaining edges of the cycle already existed when this edge was inserted, and form a path of length at most r between its endpoints. This contradicts the insertion rule.

Order the vertices of X by the time they were put aside. If $x \in X$, every later vertex of X that is a G -neighbor of x was ineligible when x was processed. All these neighbors were therefore in the radius- r ball about x in the graph $D + M$ at that time. Consequently x has at most B later neighbors in $G[X]$. Greedy coloring in reverse order gives a proper coloring of $G[X]$ with at most $B + 1$ colors. Each color class has at most m vertices, since $\alpha(G) \leq m$, and hence

$$|X| \leq m(B + 1).$$

For the weight estimate, fix a threshold $a \in [0, 1]$. Call an edge of M a crossing edge if one endpoint has weight greater than a and the other has weight at most a . Let Y_a consist of the higher-weight endpoint of each crossing edge. For every crossing edge vu , the vertex selected first by the procedure is its higher-weight endpoint v , so

$$w(v) > a \geq w(u).$$

Order Y_a by the insertion times of its matching edges. When $v \in Y_a$ was processed, every later vertex $y \in Y_a$ was still unprocessed and had weight greater than a , whereas the chosen partner u had weight at most a . If $vy \in E(G)$, then y must have been ineligible, since an eligible y would have been preferred to u . Every later G -neighbor of v in Y_a was therefore in the current radius- r ball about v . Thus every vertex of $G[Y_a]$ has at most B later neighbors. As before, $G[Y_a]$ is $(B + 1)$ -colorable, so

$$|Y_a| \leq m(B + 1).$$

The matching has exactly $|Y_a|$ crossing edges.

Integrating over thresholds gives

$$\begin{aligned} \sum_{uv \in M} |w(u) - w(v)| &= \int_0^1 |\{uv \in M : \min(w(u), w(v)) \leq a < \max(w(u), w(v))\}| da \\ &\leq m(B + 1). \end{aligned}$$

This proves (15). □

Proof of Proposition 9. Put $H = G + F$. Choose an optimal weight function w in (3) for H . Thus

$$\sum_{v \in V(G)} w(v) = \chi_f(H), \quad 0 \leq w(v) \leq 1.$$

Apply Lemma 11 to G , $D = F$, these weights, and $r = 2m - 1$. Let M be the resulting matching and let X be its unmatched set. Since $M \subseteq E(G)$,

$$Q = (G - X)/M$$

is a minor of G . Here the quotient retains every edge arising from an edge of $G - X$, removing only loops and identifying parallel edges. Assign to the quotient vertex corresponding to an edge $uv \in M$ the weight

$$z(uv) = \min\{w(u), w(v)\}.$$

We claim that z is feasible in (3) for Q . Let S be any independent set of Q , and let U be the union of its matching pairs. There is no G -edge between different pairs of S , by the definition of this quotient. Selecting one endpoint from each pair therefore gives an independent set in G , whence

$$|S| \leq m, \quad |U| \leq 2m.$$

Furthermore, $G[U]$ consists exactly of the matching $M[U]$.

No cycle of $F[U] + M[U]$ contains a matching edge: such a cycle would have length at most $|U| \leq 2m$, contrary to Lemma 11. Lemma 10 therefore gives a set $T \subseteq U$ that is independent in F and contains exactly one endpoint of every pair in S . Since $G[U]$ consists of the matching pairs, T is also independent in G , and hence in H . Therefore

$$\sum_{e \in S} z(e) \leq \sum_{v \in T} w(v) \leq 1.$$

This proves feasibility.

Lemma 11 now yields

$$\begin{aligned} 2 \sum_{e \in M} z(e) &= \sum_{v \notin X} w(v) - \sum_{uv \in M} |w(u) - w(v)| \\ &\geq \chi_f(H) - |X| - m(K + 1) \\ &\geq \chi_f(H) - 2m(K + 1). \end{aligned}$$

By fractional-coloring duality,

$$\chi_f(H) \leq 2\chi_f(Q) + 2m(K + 1).$$

Finally, (4) and $h(Q) \leq h(G)$ imply

$$\chi_f(H) \leq 4h(Q) + 2m(K + 1) \leq 4h(G) + 2m(K + 1),$$

which is (14). □

3.5 Completing the proof

Theorem 2. *For every graph G and every $0 < \varepsilon \leq 1$,*

$$\chi(G) \leq 4h(G) + \varepsilon|G| + (100/\varepsilon)^{2000/\varepsilon^2}.$$

Proof. Fix $0 < \varepsilon \leq 1$ and set

$$m = \lceil 2/\varepsilon \rceil, \quad r = m + 2, \quad \gamma = \varepsilon/6, \quad \mu = \left(\frac{\varepsilon}{2400r^2} \right)^{20r}.$$

Lemma 7 gives the pair-load tolerance $\mu/4$ and order threshold

$$n_0 = \lceil 10^6 r^4 \mu^{-4} + 12/\varepsilon \rceil. \tag{16}$$

Since $m \geq 2$, Lemma 8 applies. Every graph G on $n \geq n_0$ vertices with $\alpha(G) \leq m$ admits $F \subseteq \overline{G}$ such that

$$\Delta(F) \leq d := \left\lceil \frac{288 \binom{m}{2}}{\mu \varepsilon^2} \right\rceil, \quad \chi(G) \leq \chi_f(G + F) + \varepsilon n/2.$$

Proposition 9 now gives

$$\chi(G) \leq 4h(G) + \varepsilon n/2 + A,$$

where

$$A = \max \left\{ n_0, 2m \left(1 + \sum_{j=0}^{2m-1} (d+1)^j \right) \right\}.$$

The same inequality holds when $n < n_0$, since $\chi(G) \leq n < n_0$. Thus it holds for every graph of independence number at most m , including the empty graph.

For an arbitrary graph G on n vertices, repeatedly remove a stable set of size exactly $m+1$ while such a set exists. There are at most $n/(m+1)$ removed sets; let J be the remaining induced subgraph. Then

$$\frac{n}{m+1} \leq \frac{\varepsilon n}{2}, \quad \alpha(J) \leq m, \quad h(J) \leq h(G).$$

Color the removed sets separately and apply the preceding bound to J . This gives

$$\chi(G) \leq \frac{n}{m+1} + \chi(J) \leq \frac{\varepsilon n}{2} + 4h(J) + \frac{\varepsilon |V(J)|}{2} + A \leq 4h(G) + \varepsilon n + A.$$

Finally, we bound that fully explicit expression. Put $R = 100/\varepsilon$. Since $m \leq 3/\varepsilon$ and $r \leq 5/\varepsilon$,

$$\mu^{-1} = \left(\frac{2400r^2}{\varepsilon} \right)^{20r} \leq (60000\varepsilon^{-3})^{100/\varepsilon} \leq R^{300/\varepsilon}.$$

The last step uses $60000\varepsilon^{-3} \leq R^3$. Also, $\binom{m}{2} \leq 9/(2\varepsilon^2)$, so the ceiling in the definition of d yields

$$d+1 \leq \frac{1298}{\varepsilon^4} \mu^{-1} \leq R^{304/\varepsilon}.$$

Here $1298\varepsilon^{-4} \leq R^{4/\varepsilon}$. Using $\sum_{j=0}^{2m-1} (d+1)^j \leq 2m(d+1)^{2m}$ and $2m \leq 6/\varepsilon$ gives

$$2m \left(1 + \sum_{j=0}^{2m-1} (d+1)^j \right) \leq \left(\frac{6}{\varepsilon} + \frac{36}{\varepsilon^2} \right) (d+1)^{2m} \leq 42\varepsilon^{-2} R^{1824/\varepsilon^2} \leq R^{1826/\varepsilon^2}.$$

The final step uses $42\varepsilon^{-2} \leq R^{2/\varepsilon^2}$. For the other term in A , (16) and $\varepsilon \leq 1$ give

$$n_0 \leq 10^6 (5/\varepsilon)^4 R^{1200/\varepsilon} + 12/\varepsilon + 1 \leq (6.25 \cdot 10^8 \varepsilon^{-4} + 13\varepsilon^{-1}) R^{1200/\varepsilon} \leq R^{1206/\varepsilon^2}.$$

Therefore $A \leq R^{2000/\varepsilon^2}$, which proves (1). □

4 Proof of the exponent improvement

We prove Theorem 3. Packing connected induced bipartite subgraphs produces a smaller minor, which is colored using the assumed local bound. In the remainder, a short induced path localizes chromatic number whenever separation fails. The same local bound then supplies the separation needed for a recursive minor construction. The construction follows Delcourt and Postle, with the chromatic number of its routing subgraph controlling the cost of deletion.

4.1 Packing and localization

Let $b(G)$ be the largest order of a connected induced bipartite subgraph of G , with $b(\emptyset) = 0$. This parameter is inherited by induced subgraphs.

The first reduction separates a union of bipartite subgraphs from a remainder in which b is small.

Lemma 12. *For every graph G and integer $k \geq 2$, there is a partition $V(G) = W \dot{\cup} R$ and a minor Q of G such that*

$$|Q| \leq \lfloor |G|/k \rfloor, \quad b(G[R]) < k, \quad \chi(G[W]) \leq 2\chi(Q).$$

Proof. Greedily choose disjoint sets of k vertices inducing connected bipartite graphs, until no more can be chosen. Every connected graph of order at least k has a connected induced subgraph of order exactly k : delete leaves of a spanning tree until k vertices remain. Thus the remaining induced graph has $b < k$. Let W be the union of the chosen sets. Contract each chosen set in $G[W]$ to obtain Q , retaining all adjacencies between different sets. A coloring of Q , together with a separate bipartition bit inside each contracted set, is a coloring of $G[W]$ with at most $2\chi(Q)$ colors. The empty packing is allowed. $\square$

When chromatic separation fails, the next lemma localizes the excess chromatic number in the closed neighborhood of a short induced path.

Lemma 13. *Let $q \geq 1$ and $k \geq 2$ be integers. Suppose that $b(G) \leq k$, $\chi(G) \geq q$, and G has no two vertex-disjoint subgraphs of chromatic number at least q . Then there is an induced path P of order at most k such that, with $J = G[N_G[V(P)]]$,*

$$\chi(G - V(J)) \leq q - 1, \quad \alpha(J) \leq k(k - 1), \quad \chi(J) \geq \chi(G) - q + 1.$$

Here $N_G[X]$ denotes X together with all vertices having a neighbor in X .

Proof. At most one component of any induced subgraph of G has chromatic number at least q , since two such components would violate the hypothesis. Let C_0 be the unique such component of G , and choose $v_1 \in V(C_0)$.

We grow an induced path $P_i = v_1 \cdots v_i$. If all components of $G - N_G[V(P_i)]$ have chromatic number less than q , we stop. Otherwise let C_i be its unique component of chromatic number at least q . The construction will ensure that C_i is contained in C_{i-1} and that v_i has a neighbor in C_{i-1} when $i \geq 2$. For $i = 1$, connectivity of C_0 supplies a vertex $v_2 \in N_G(v_1)$ adjacent to C_1 . For $i \geq 2$, the component C_i is a component of $C_{i-1} - N_G(v_i)$. It is a proper subset, because v_i has a neighbor in C_{i-1} . Since C_{i-1} is connected, there is a vertex

$$v_{i+1} \in V(C_{i-1}) \cap N_G(v_i)$$

adjacent to C_i . This vertex lies outside the closed neighborhoods of $v_1, \dots, v_{i-1}$, so the extended path is induced. Every component discarded earlier had chromatic number less than q ; deleting further vertices cannot increase that number. Hence any surviving component of chromatic number at least q lies in C_i , establishing the invariant at the next step.

An induced path is bipartite, so the process cannot extend to $k + 1$ vertices. At termination its closed neighborhood therefore has the required complement. Also $\alpha(G[N_G[v]]) \leq k - 1$ for every vertex v : a stable set of k neighbors would form an induced star of order $k + 1$, and a stable set in the closed neighborhood containing v has order one. Taking the union of the closed neighborhoods of the at most k path vertices gives $\alpha(J) \leq k(k - 1)$. Finally $\chi(G) \leq \chi(J) + \chi(G - V(J))$ gives the last inequality. $\square$

The fractional bound (4) converts the independence bound into an order bound. If (x_S) is an optimal fractional coloring of H , summing its covering constraints gives

$$|H| \leq \sum_S |S| x_S \leq \alpha(H) \sum_S x_S = \alpha(H) \chi_f(H) \leq 2\alpha(H) h(H). \quad (17)$$

This lets us turn a local coloring bound into chromatic separation.

Lemma 14. *Suppose every K_u -minor-free graph of order at most $2uk^2$ has chromatic number at most du , where u, d, k are positive integers and $k \geq 2$. If Y is K_u -minor-free, $b(Y) \leq k$, and $\chi(Y) > du$, then Y has two vertex-disjoint induced subgraphs, each of chromatic number at least $\chi(Y) - du$.*

Proof. If the asserted pair does not exist, apply Lemma 13 with $q = \chi(Y) - du \geq 1$. The resulting induced graph J satisfies

$$\alpha(J) \leq k(k-1), \quad \chi(J) \geq \chi(Y) - (\chi(Y) - du) + 1 = du + 1.$$

It is K_u -minor-free, so (17) gives

$$|J| \leq 2\alpha(J)h(J) \leq 2k(k-1)(u-1) < 2uk^2.$$

This contradicts the assumed local bound. □

4.2 Connectivity tools and a hub of bounded chromatic number

We state the results needed for the recursive construction. All references to Delcourt and Postle use the numbering in [1]. Graphs are finite and simple, $|H| = |V(H)|$, and logarithms are natural. A graph is r -connected if it has more than r vertices and remains connected after deleting fewer than r vertices. We denote its vertex connectivity by $\kappa(H)$ and its minimum degree by $\delta(H)$. A K_r model consists of r pairwise disjoint nonempty connected branch sets, every two of which are joined by an edge. It is rooted at r specified distinct vertices if each branch set contains its corresponding root. A linkage is a family of pairwise vertex-disjoint paths joining prescribed pairs. A path from a set U to a disjoint set V meets these sets only at its corresponding ends. For a path family $\mathcal{P}$ or a model $\mathcal{M}$, write $V(\mathcal{P})$ or $V(\mathcal{M})$ for the union of its vertex sets.

The first theorem restores connectivity after vertices have been deleted.

Theorem 15 (Girão–Narayanan, [1, Theorem 5.1]). *Let $r \geq 1$ be an integer. Every graph H with $\chi(H) \geq 7r$ contains an r -connected subgraph H' with $\chi(H') \geq \chi(H) - 6r$.*

We may take the induced closure of H' , since this preserves connectivity and cannot decrease its chromatic number.

We use Menger's theorem and the following quantitative linkedness result.

Theorem 16 (Thomas–Wollan, [1, Theorem 5.6 and the following remark]). *Let $\ell \geq 1$ be an integer. Every 10ℓ -connected graph contains a linkage joining any prescribed ℓ pairs of distinct terminals.*

A sufficiently large complete minor also supplies prescribed roots.

Lemma 17 (Kawarabayashi, [1, Lemma 5.14]). *Let $r \geq 1$ be an integer. Every r -connected graph containing a K_{2r} minor has a K_r model rooted at any r specified distinct vertices.*

For the initial scale we use the density theorem and its coloring consequence (2), stated before the proof of Theorem 1.

The next lemma supplies a hub whose deletion has bounded chromatic cost.

Lemma 18. *Let $r \geq 1$ be an integer. If $\chi(H) \geq 7r$, then H has an induced r -connected subgraph H_0 with $\chi(H_0) \leq 7r$.*

Proof. Delete vertices one at a time to obtain an induced subgraph X with $\chi(X) = 7r$. Apply Theorem 15 to X and take the induced closure of the resulting subgraph. $\square$

To join the hub to both the prescribed terminals and the remaining graph, we use Delcourt and Postle's redundancy lemma.

Lemma 19 (Delcourt–Postle, [1, Lemma 5.3]). *Let A_1, A_2, B be disjoint nonempty vertex sets in a graph H . For each $i \in \{1, 2\}$, suppose that $H - A_{3-i}$ contains $2|A_i|$ paths from A_i to B , pairwise disjoint outside A_i , with exactly two paths starting at each vertex of A_i . Then H contains $|A_1| + |A_2|$ pairwise vertex-disjoint paths from $A_1 \cup A_2$ to B .*

We record the consequence used below.

Lemma 20. *Let $a \geq 1$ be an integer, and let Z, U, V be disjoint vertex sets in a graph, with $|Z| = 7a$. Suppose there are $14a$ paths from Z to V , two from each vertex of Z , disjoint outside Z and avoiding U , and $2a$ disjoint paths from U to V avoiding Z . Then there are $8a$ disjoint paths to V , one from each vertex of Z and a from distinct vertices of U , with all interiors outside $Z \cup U \cup V$.*

Proof. Work in the subgraph consisting of the given paths and their edges. Pair the $2a$ starting vertices in U of the second path family. For each pair add a new vertex adjacent to its two members, and let A_2 be the set of the a new vertices. Extending the second path family by these edges gives two paths from each vertex of A_2 to V , disjoint outside A_2 and avoiding Z . The first family avoids A_2 . Lemma 19, with $A_1 = Z$ and $B = V$, gives $8a$ disjoint paths, one from each vertex of $Z \cup A_2$.

Delete the new starting vertices. Vertices of U have degree at most one in the union of the original paths, so they cannot be internal vertices of the resulting paths. The other exclusions hold by construction. $\square$

We use the rooted convention of [1, Definition 5.12]. For integers $a \geq 1$ and $b \geq 0$, a graph is (a, b) -woven if, for any distinct roots $r_1, \dots, r_a$ and terminal pairs $(s_1, t_1), \dots, (s_b, t_b)$, with different pairs having no common terminal but allowing $s_i = t_i$, it contains a K_a model rooted at the roots and a linkage for the pairs whose vertex intersection is exactly the roots that are also terminals. A singleton pair is represented by a path of length zero. Roots may coincide with terminals.

The following form of the weaving lemma includes the containment properties established in its cited proof.

Lemma 21 (Delcourt–Postle, [1, Lemma 5.15 and its proof]). *Let $a, b \geq 1$ be integers. Let H be an (a, b) -woven subgraph of a graph F , with $|H| \geq a + 2b$, and let $R \subseteq V(H)$ have size a . If $\mathcal{P}$ is a linkage in F for b pairs with all $2b$ terminals distinct, then there are a K_a model $\mathcal{M}$ in H rooted at R and a linkage $\mathcal{P}'$ for the same pairs such that*

$$V(\mathcal{P}') \subseteq V(\mathcal{P}) \cup V(H), \quad V(\mathcal{P}') \cap V(\mathcal{M}) \subseteq R \cap V(\mathcal{P}).$$

The following simultaneous form follows from Delcourt and Postle's linkage rerouting argument [1, Claim 7.1.1].

Lemma 22. *Let a be a positive integer divisible by three, and put $a' = 2a/3$. Suppose that a graph F contains disjoint $(a', 2a)$ -woven subgraphs J_1, J_2, J_3 , each of order at least $a' + 4a$. Let $s_1, \dots, s_{2a}$ be distinct vertices outside their union, and let $t_1, \dots, t_{2a}$ be distinct vertices such that*

$$T_i = \{t_{(i-1)a'+1}, \dots, t_{ia'}\} \subseteq V(J_i) \quad (i = 1, 2, 3).$$

If F has a linkage for the pairs (s_j, t_j) , then it has a linkage $\mathcal{P}$ for the same pairs and $K_{a'}$ models $\mathcal{M}_i$ in J_i , rooted at T_i , such that $V(\mathcal{P}) \cap V(\mathcal{M}_i) = T_i$ for $i = 1, 2, 3$.

Proof. Starting with the given linkage, apply Lemma 21 to J_1, J_2, J_3 in that order, always with all $2a$ terminal pairs. At each application, the linkage meets the new model only at roots that were already on the linkage. All a' roots in T_i are endpoints of its paths, so the intersection is exactly T_i . The new linkage lies in the union of the previous one and J_i . Since the three subgraphs are disjoint, rerouting in J_i cannot create an intersection with either of the models constructed earlier. Their roots remain on the linkage as endpoints. $\square$

4.3 The outer recursion with a separation hypothesis

For a nonnegative integer s , a graph Y is s -chromatic-separable if it has two vertex-disjoint subgraphs, each of chromatic number at least $\chi(Y) - s$ [1, Definition 2.4]. We may require these subgraphs to be induced by taking their induced closures.

The next lemma adapts the proof of [1, Theorem 7.1]. Here separation is a hypothesis, and Lemma 18 bounds the chromatic cost of the hub. The linkage and weaving steps are supplied by the results stated above.

Lemma 23. *Let $T \geq 100$ be a power of three and let $d \geq 1$ be an integer. Suppose that a graph G has the following property: for every integer scale $a = (2/3)^i T > T/\sqrt{\log T}$, where $i \geq 0$, every induced K_{14a} -minor-free subgraph Y of G with $\chi(Y) > 28da$ is $14da$ -chromatic-separable. Put*

$$K = 10000, \quad A = 2000, \quad B = 10^6(d + 1).$$

For every integer scale $a = (2/3)^i T$ with $i \geq 0$, each induced subgraph F of G with $\kappa(F) \geq Ka$ and $\chi(F) \geq AT + Ba$ is $(a, 3a)$ -woven.

Proof. We induct on a among the integer scales. We first separate any coincident roots and terminals. Let Z_0 be the set of vertices occupying these roles, so $|Z_0| \leq 7a$. Greedily choose a distinct neighbor outside Z_0 for each of the $7a$ roles, including both roles of a singleton pair; this is possible since $\delta(F) > 14a$. Call these new vertices the proxies, and put $F_0 = F - Z_0$. Then

$$\kappa(F_0) \geq (K - 7)a, \quad \chi(F_0) \geq AT + (B - 7)a.$$

A model and linkage for the distinct proxies extend to the original vertices by their chosen edges. For a singleton pair, discard its proxy path and use its original vertex. Since F_0 avoids Z_0 , this gives exactly the permitted root-terminal intersections.

If F_0 contains a K_{14a} minor, Lemma 17 gives a K_{7a} model rooted at all the proxies, since $\kappa(F_0) \geq 7a$. Retain its a root branches, and join each pair of terminal branches by an edge and paths inside those branches. This gives the required model and linkage. In particular, this applies when $a \leq T/\sqrt{\log T}$, because

$$840a\sqrt{\log(14a)} + 1 \leq 840T\sqrt{\frac{\log(14T)}{\log T}} + 1 \leq 840T\sqrt{1 + \frac{\log 14}{\log 100}} + 1 < 2000T$$

for $T \geq 100$, and (2) forces a K_{14a} minor in F_0 . We may therefore assume that $a > T/\sqrt{\log T}$ and that F_0 is K_{14a} -minor-free.

Linking through a hub. Let Z be the set of $7a$ proxies. Apply Lemma 18 in $F_0 - Z$ with $r = 140a$ to obtain an induced subgraph H_0 with

$$\kappa(H_0) \geq 140a, \quad \chi(H_0) \leq 980a.$$

The hypothesis holds because $\chi(F_0 - Z) \geq AT + (B - 14)a \geq 980a$. Since $\kappa(F_0) \geq (K - 7)a \geq 14a$ and $|H_0| \geq 14a$, Menger's theorem, with each source in Z replaced by two copies, gives $14a$ paths from Z to H_0 , two from each source and disjoint outside Z . Choose each path induced by shortcutting it. The graph induced by their union has chromatic number at most $28a$, using a separate two-color palette for each path.

Delete Z , H_0 , and these paths from F_0 , and apply Theorem 15 with $r = Ka$ to the remaining induced graph. Its chromatic number is at least $AT + (B - 1022)a > 7Ka$. We obtain an induced subgraph F_3 with

$$\kappa(F_3) \geq Ka, \quad \chi(F_3) \geq AT + (B - 1022 - 6K)a.$$

It suffices to prove that F_3 is $(a, 0)$ -woven. To see this, apply Menger's theorem in $F_0 - Z$ to find $2a$ disjoint paths from F_3 to H_0 ; the host is at least $2a$ -connected because $\kappa(F_0 - Z) \geq (K - 14)a \geq 2a$, and both endpoint sets have size at least $2a$. The earlier double fan avoids F_3 , so Lemma 20 gives $8a$ disjoint paths to H_0 , one from each proxy and a from distinct vertices of F_3 , with interiors outside these endpoint sets. Root a K_a model in F_3 at the latter ends. By Theorem 16, the hub has a linkage pairing its $8a$ ends as required: $3a$ pairs connect the terminal paths, and a pairs connect the root paths to the paths from F_3 . This applies since $\kappa(H_0) \geq 140a \geq 10(4a)$. The unions give the required model and linkage in F_0 .

Separating and applying induction. Fix distinct roots $r'_1, \dots, r'_a$ in F_3 . Since $\delta(F_3) \geq Ka$, choose distinct vertices $s_1, \dots, s_{2a}$ outside the roots so that s_j, s_{a+j} are adjacent to r'_j . Delete these $3a$ vertices to obtain F_4 . Apply the separation hypothesis to F_4 and then to one of the two resulting induced subgraphs, obtaining three disjoint induced subgraphs J_1, J_2, J_3 . All these graphs are K_{14a} -minor-free. The two applications are valid because

$$\chi(F_4) \geq AT + (B - 1025 - 6K)a, \quad B - 1025 - 6K - 14d > 28d.$$

The first separation costs at most $14da$, so the input to the second has chromatic number at least $AT + (B - 1025 - 6K - 14d)a > 28da$. A second loss of at most $14da$ leaves each of the three graphs with $\chi(J_i) \geq AT + (B - 1025 - 6K - 28d)a > 7Ka$. Apply Theorem 15 inside each J_i with $r = Ka$ and take induced closures to obtain J'_1, J'_2, J'_3 .

The total chromatic loss for each child is at most

$$\underbrace{7a + 7a}_{\text{original vertices and proxies}} + \underbrace{980a + 28a}_{\text{hub and paths}} + \underbrace{12Ka}_{\text{connectivity}} + \underbrace{3a}_{\text{roots and neighbors}} + \underbrace{28da}_{\text{separation}}.$$

Adding these costs gives $7 + 7 + 980 + 28 + 3 + 28d + 12K = 1025 + 28d + 12K$. Consequently

$$\kappa(J'_i) \geq Ka, \quad \chi(J'_i) \geq AT + (B - 1025 - 28d - 12K)a \geq AT + B(2a/3), \quad (18)$$

where the last inequality follows from $B/3 > 1025 + 28d + 12K$. The next scale $a' = 2a/3$ is integral: if $T = 3^m$, then the integer scales are $(2/3)^i T$ for $0 \leq i \leq m$, and the last is $2^m \leq 3^m / \sqrt{m \log 3} = T / \sqrt{\log T}$. This inequality holds for $m \geq 5$ (check $m = 5$, then use that

$(2/3)^m \sqrt{m \log 3}$ decreases). Thus every nonbase scale occurs before the last integer scale. Induction and (18) imply that each J'_i is $(a', 3a') = (a', 2a)$ -woven.

Assembling the rooted model. Choose a' distinct targets in each child, indexed as $t_1, \dots, t_{2a}$ in three consecutive blocks of length a' . The graph $F_3 - \{r'_1, \dots, r'_a\}$ has connectivity at least $(K-1)a \geq 20a$. Theorem 16 therefore supplies a linkage from s_j to t_j for $1 \leq j \leq 2a$. Since $|J'_i| \geq Ka + 1 \geq a' + 4a$, Lemma 22 supplies a linkage for the same pairs and rooted $K_{a'}$ models in the children, meeting the linkage exactly at their targets.

For each $j \in \{1, \dots, a\}$, unite r'_j , its two chosen neighbor edges, the paths indexed j and $a+j$, and the child branches containing t_j and t_{a+j} . These sets are disjoint and connected. Each uses two different children, because the two target indices differ by a , exceeding the child block length $2a/3$. Any two two-element subsets of the three children intersect, and the corresponding distinct branches in a common child are adjacent. The resulting sets form a K_a model rooted at the chosen roots. Thus F_3 is $(a, 0)$ -woven, completing the induction. $\square$

Corollary 24. *Let $t \geq 100$ be an integer, and let T be the least power of three with $T \geq t$. If a K_t -minor-free graph G satisfies the separation hypothesis of Lemma 23 for this T and an integer $d \geq 1$, then*

$$\chi(G) < 3(10^6(d+1) + 62000)t.$$

Proof. Let A, B, K be the constants in Lemma 23. If $\chi(G) \geq (A+B+6K)T$, Theorem 15 gives an induced KT -connected subgraph F with $\chi(F) \geq (A+B)T$; its hypothesis holds since $A+B+6K \geq 7K$. Lemma 23, at scale $a = T$, gives a K_T minor, contradicting K_t -minor-freeness. Now use $T < 3t$ and $A+6K = 62000$. $\square$

4.4 Completing the proof

Theorem 3. *If a local linear bound holds at exponent $\alpha > 0$, then a local linear bound holds at exponent $4\alpha/3$.*

Proof. Let D, t_0 be constants for the bound at exponent α , and increase D to an integer. For sufficiently large t , let G be K_t -minor-free with $|G| \leq t(\log t)^{4\alpha/3}$, and put

$$k = \lceil (\log t)^{\alpha/3} \rceil \geq 2.$$

Lemma 12 gives W, R, Q with

$$|Q| \leq \frac{|G|}{k} \leq \frac{t(\log t)^{4\alpha/3}}{(\log t)^{\alpha/3}} = t(\log t)^\alpha.$$

Since Q is K_t -minor-free, the bound at exponent α gives $\chi(G[W]) \leq 2Dt$.

Write $H = G[R]$. We have $b(H) < k$. Let T be the least power of three at least t . At every nonbase scale of Lemma 23, put $u = 14a$. Uniformly throughout these scales,

$$\frac{14T}{\sqrt{\log T}} < u \leq 14T, \quad \log u > \log(14T) - \frac{1}{2} \log \log T \geq \frac{1}{2} \log t$$

for all sufficiently large t . Moreover, $k \leq 2(\log t)^{\alpha/3}$, so

$$2k^2 \leq 8(\log t)^{2\alpha/3} \leq \left(\frac{1}{2} \log t\right)^\alpha \leq (\log u)^\alpha.$$

The middle inequality holds eventually because $\alpha/3 > 0$. Increasing the lower threshold for t if needed, we have, uniformly in this window,

$$u \geq t_0.$$

The bound at exponent α therefore colors every K_u -minor-free graph on at most $2uk^2$ vertices with at most Du colors. Now let Y be an induced K_u -minor-free subgraph of H with $\chi(Y) > 28Da = 2Du$. Since $b(Y) \leq b(H) < k$, we can apply Lemma 14 to Y . It gives two disjoint induced subgraphs with chromatic number at least $\chi(Y) - Du = \chi(Y) - 14Da$. This verifies the separation hypothesis of Lemma 23 with $d = D$ at every nonbase scale.

Corollary 24 now gives

$$\chi(H) < 3(10^6(D + 1) + 62000)t.$$

Color W and R with disjoint palettes. We obtain

$$\chi(G) \leq \left\lceil 2D + 3(10^6(D + 1) + 62000) \right\rceil t.$$

Both uses of a local coloring bound, on Q and on the small neighborhood cores, used exponent α . This proves Theorem 3. $\square$

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